

Problem

Does there exist a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 + f(y)) = f(x)^2 - y \quad \forall x, y \in \mathbb{R}?$$

Answer

Assume such a function exists.

Substitute $x = 0$:

$$f(f(y)) = f(0)^2 - y \quad \forall y \in \mathbb{R}.$$

Hence the right-hand side is bijective in y , so f is bijective.

Now substitute $y = 0$:

$$f(x^2 + f(0)) = f(x)^2 \quad \forall x \in \mathbb{R}.$$

Replacing x by $-x$ gives:

$$f(x^2 + f(0)) = f(-x)^2.$$

Thus,

$$f(x)^2 = f(-x)^2.$$

Since f is injective, this implies

$$f(-x) = -f(x) \quad \forall x \neq 0.$$

Since f is surjective, there exists $r \in \mathbb{R}$ such that $f(r) = 0$. If $r \neq 0$, then

$$f(-r) = -f(r) = 0 = f(r),$$

contradicting injectivity. Hence $r = 0$, so $f(0) = 0$.

Substitute $(x, y) = (1, 0)$:

$$f(1) = f(1)^2 \Rightarrow f(1) \in \{0, 1\}.$$

Since $f(0) = 0$ and f is injective, $f(1) \neq 0$, so $f(1) = 1$.

Finally substitute $(x, y) = (0, 1)$:

$$f(f(1)) = f(0)^2 - 1 = -1.$$

But $f(1) = 1$, so

$$f(f(1)) = f(1) = 1,$$

giving $1 = -1$, a contradiction.

Therefore, no such function exists.